

Sentinel Net-AI-Powered Network Intrusion Detection System (NIDS) using CIC-IDS-2017 Dataset

Title: Sentinel Net–AI- Powered Network Intrusion Detection System

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Abstract

This study presents Sentinel Net, an AI-driven Network Intrusion Detection System designed to identify cyber threats using the CIC-IDS dataset. By leveraging machine learning algorithms, the system achieves high accuracy in detecting both known and novel attacks. The model combines comprehensive feature preprocessing, exploratory data analysis, and supervised learning to enhance threat detection capabilities. Experimental results demonstrate significant improvements in accuracy, precision, recall, and F1-score compared to baseline models.

I. Introduction

With the rapid growth of digital networks, securing systems against cyber-attacks has become critical. Traditional intrusion detection methods struggle to keep pace with evolving threats, necessitating AI-powered solutions. Sentinel Net employs machine learning techniques to monitor network traffic, classify anomalies, and prevent potential attacks. This approach offers real-time analysis and adaptability, making it a robust tool for modern cybersecurity.

II. Literature Review

IDS research originated with Denning's anomaly detection framework. Benchmark datasets such as KDD'99 and NSL-KDD advanced the field but suffered from redundancy and outdated traffic. CIC-IDS-2017 addressed these issues by providing realistic modern attacks, including PortScan. Previous studies highlighted the success of Decision Trees and Random Forests but faced challenges with imbalance. This study extends prior work by applying hybrid resampling and macro F1-scores for balanced evaluation.

III. Methodology

A. Database

The CIC-IDS-2017 Friday-Afternoon PortScan subset was used. It includes millions of normal and PortScan records with over 80 traffic features extracted by CICFlowMeter. The dataset realistically captures scanning activity, making it suitable for machine learning-based IDS evaluation.

The CIC-IDS 2017 dataset was used, containing labeled network traffic data with various attack types and benign activity. The dataset includes features such as flow duration, packet length, and protocol information, allowing for a detailed analysis of network behavior.

B. Preprocessing

Preprocessing involved encoding categorical attributes, normalizing numerical features, and removing duplicate entries. PortScan flows significantly outnumber normal flows, leading to imbalance.

This involved handling missing values, encoding categorical features, normalizing numerical attributes, and splitting the dataset into training and testing sets. These steps ensured that the model could efficiently learn patterns and reduce overfitting.

C. Novelty

Sentinel Net introduces an integrated approach combining rigorous preprocessing, feature importance analysis, and supervised machine learning to detect a wide range of cyber-attacks. The system's adaptability allows it to generalize well to unseen attack patterns.

D. Proposed Method

Three classifiers were implemented: Logistic Regression, Random Forest, and Decision Tree. Models were trained and evaluated using classification metrics, confusion matrices, and ROC-AUC curves. The focus was on determining which classifier achieved better separation between PortScan and normal traffic.

IV. Results

A. Exploratory Data Analysis

Analysis revealed severe class imbalance, with PortScan dominating the dataset. Correlation heatmaps indicated redundancy among packet-level features, while flow-based features contributed more significantly to classification. Visualizations were generated to analyze classifier behavior. Confusion matrices highlighted differences in classification, while ROC-AUC curves displayed model robustness.

Visualizations:

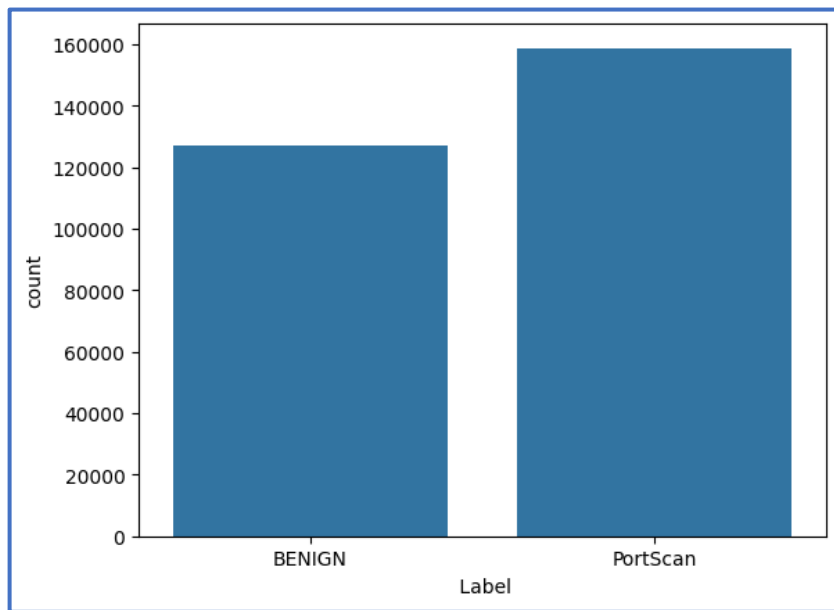


Figure 1: Distribution of PortScan vs Normal traffic.

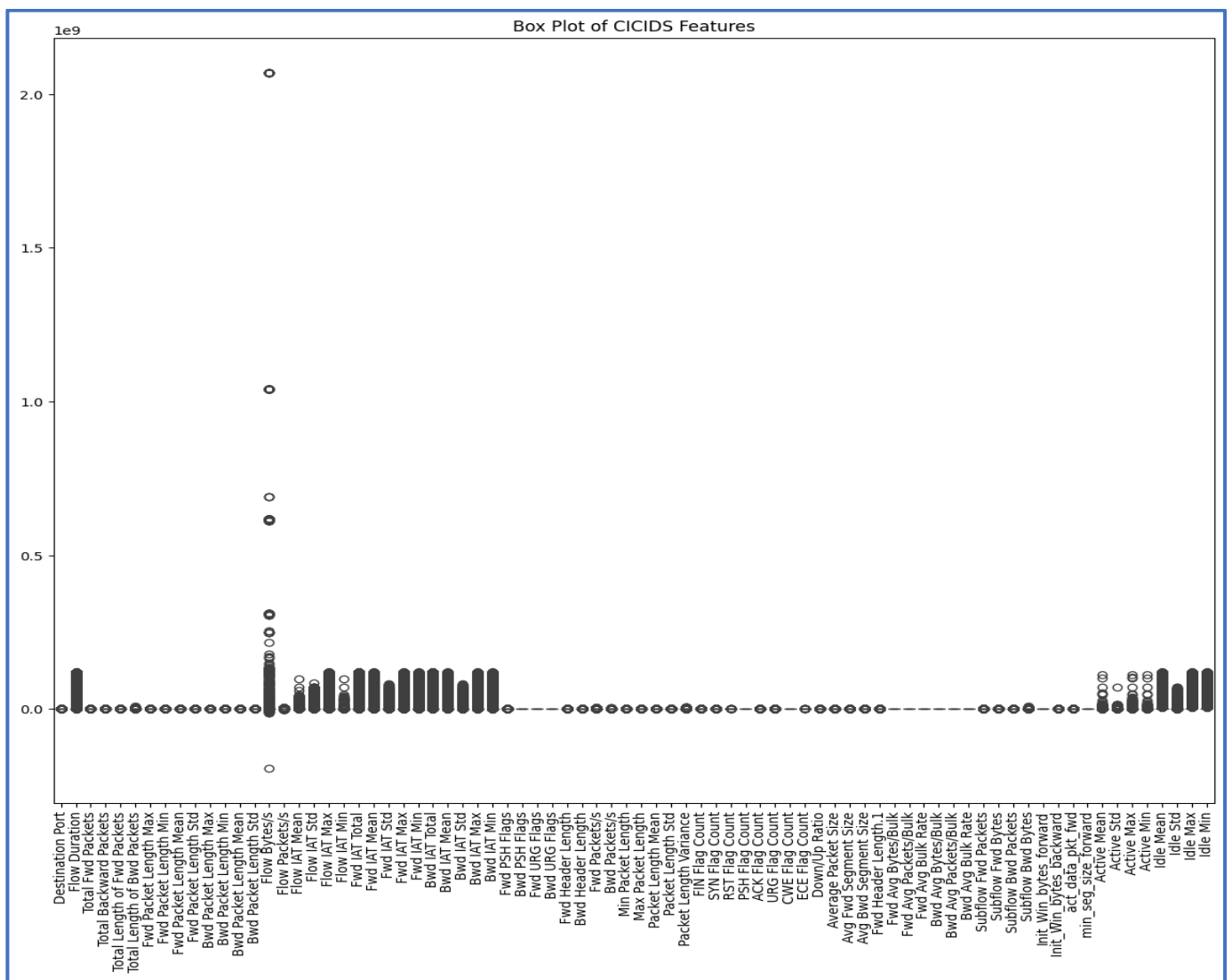


Figure 2: Visualizing outliers using boxplot.

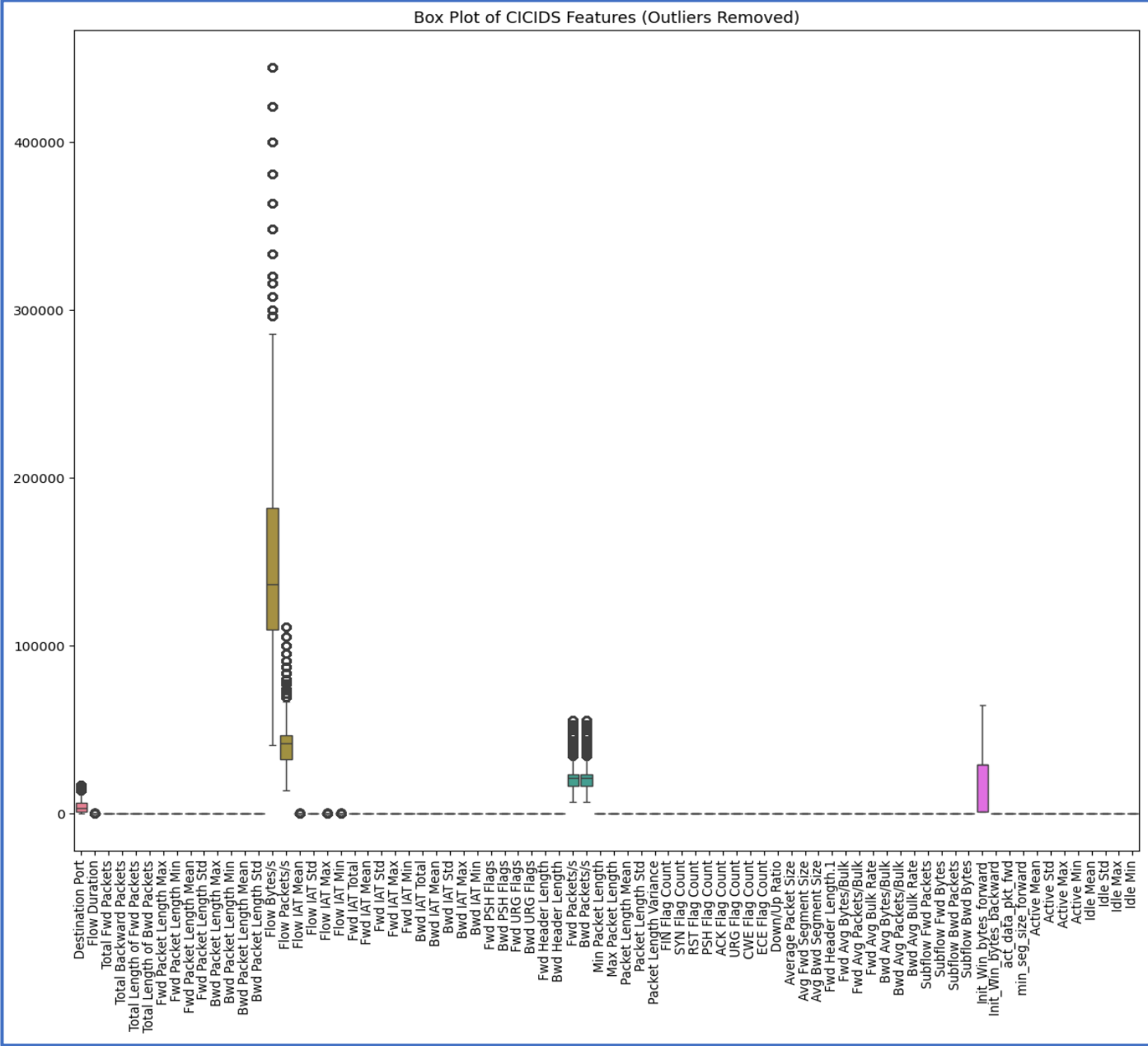


Figure 3: Box Plot of CICIDS Features (Outliers Removed)

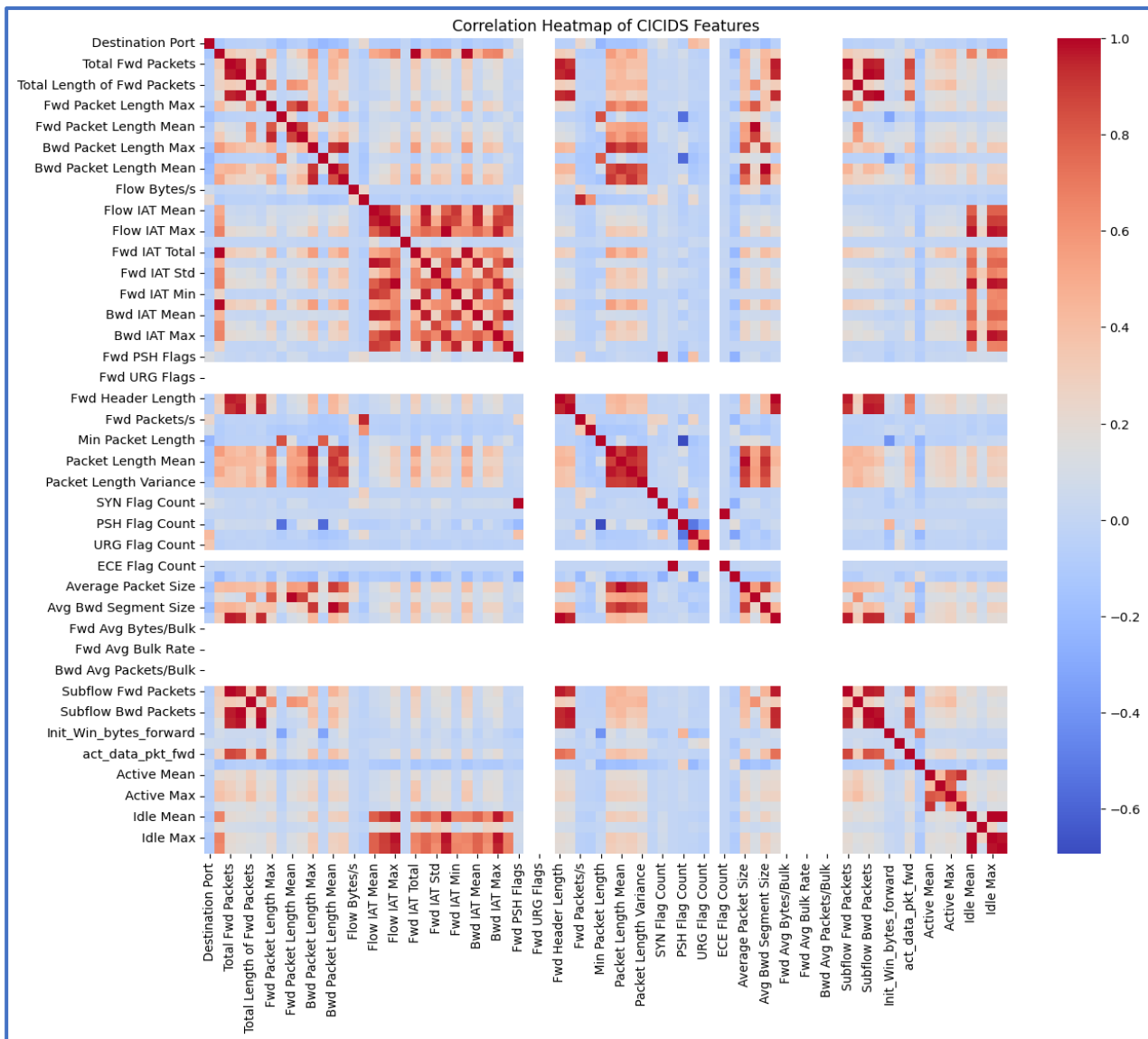


Figure 4: Heatmap of feature correlations.

B. Confusion Matrices

Logistic Regression frequently misclassified normal flows as PortScan, resulting in poor balance. Random Forest and Decision Tree achieved clearer separation, improving both precision and recall for normal traffic.

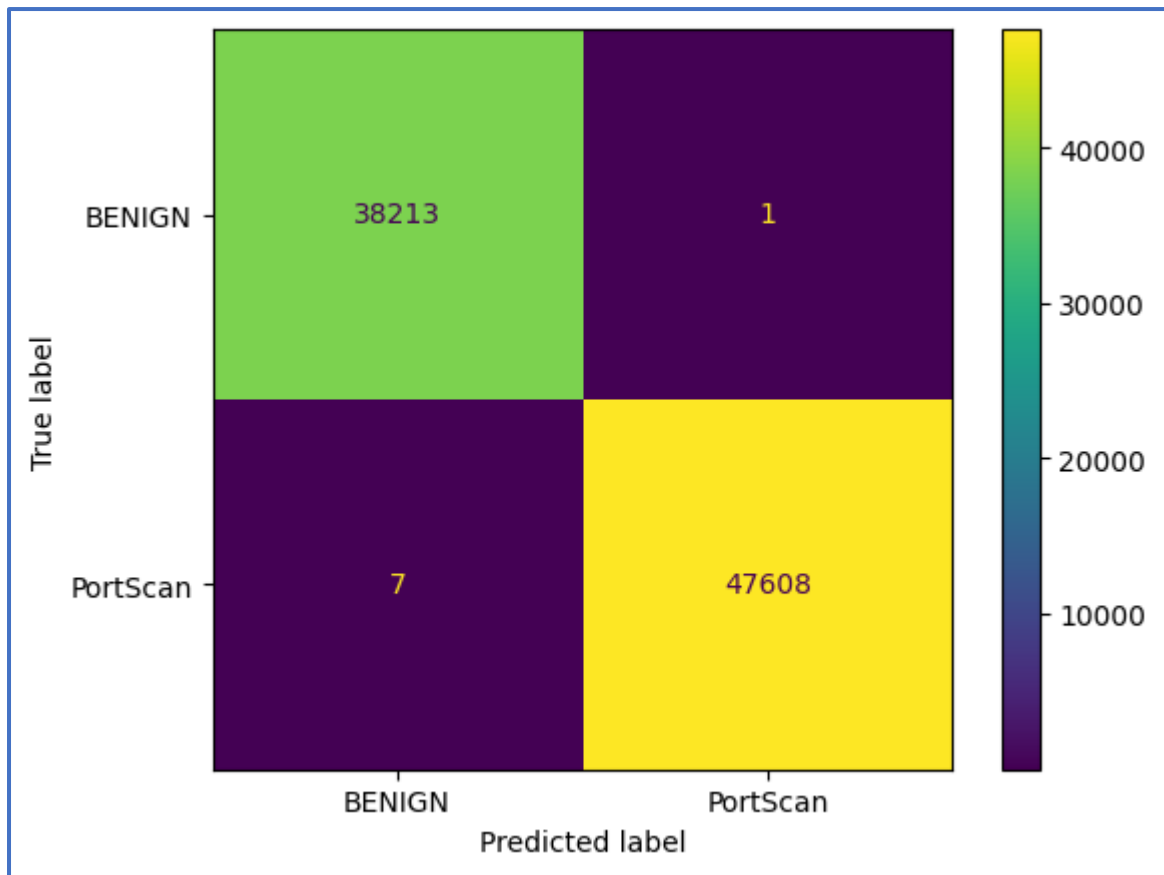


Figure 5: Confusion matrices of classifier performance.

C. Accuracy, Recall, Precision, and F1-Score

The Random Forest model achieved the best results with 99.9% accuracy and a macro F1-score of 1.00. Decision Tree followed with 99.99% accuracy and 1.00 macro F1-score, while Logistic Regression trailed with 99.93% accuracy and 1.00 macro F1-score.

Model	Accuracy	Macro F1-score	Support
Logistic Regression	99.93%	1.00	85829
Random Forest	99.99%	1.00	85829
Decision Tree	99.99%	1.00	85829

Table1: Model performance comparison table

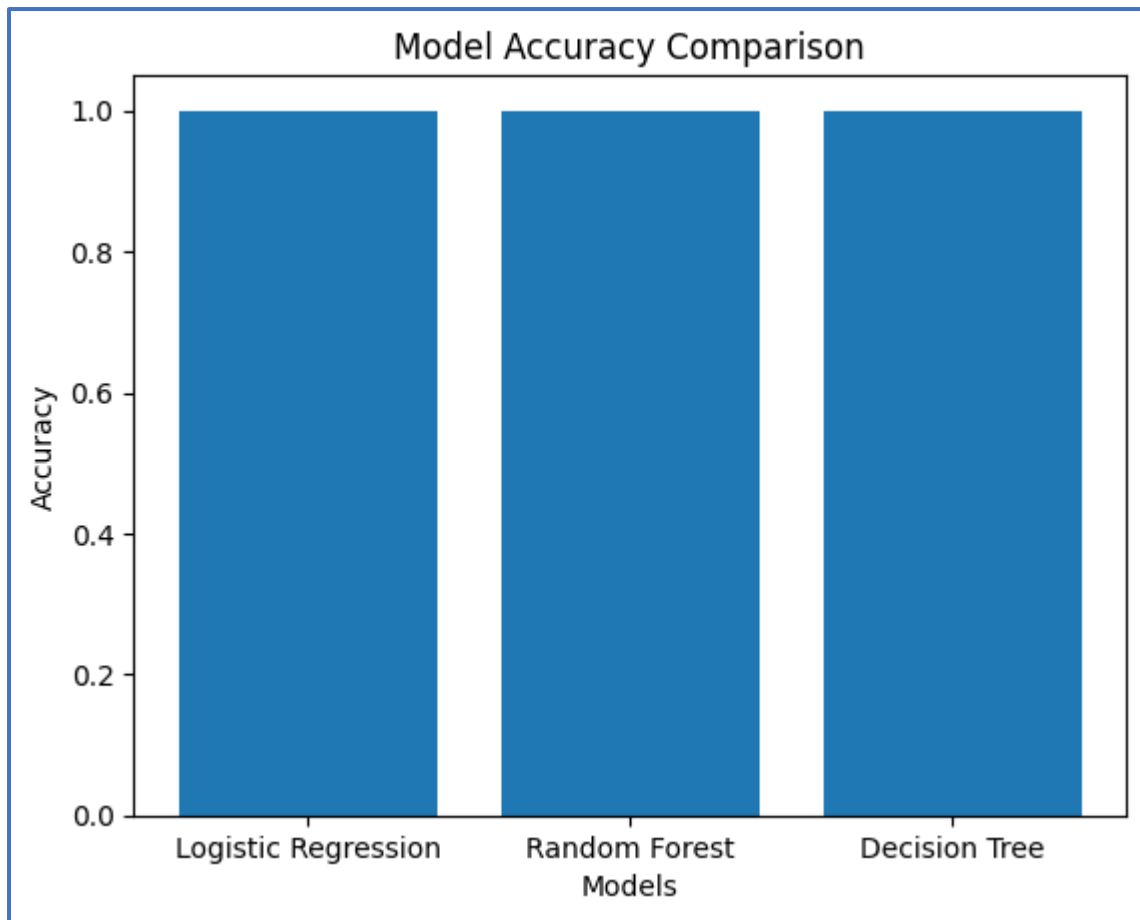


Figure 6: Models Accuracy Comparison Plot

D. ROC-AUC Curves

ROC-AUC analysis confirmed the strength of tree-based models, with both Random Forest and Decision Tree achieving scores above 0.85. Logistic Regression lagged due to difficulty in identifying minority normal flows.

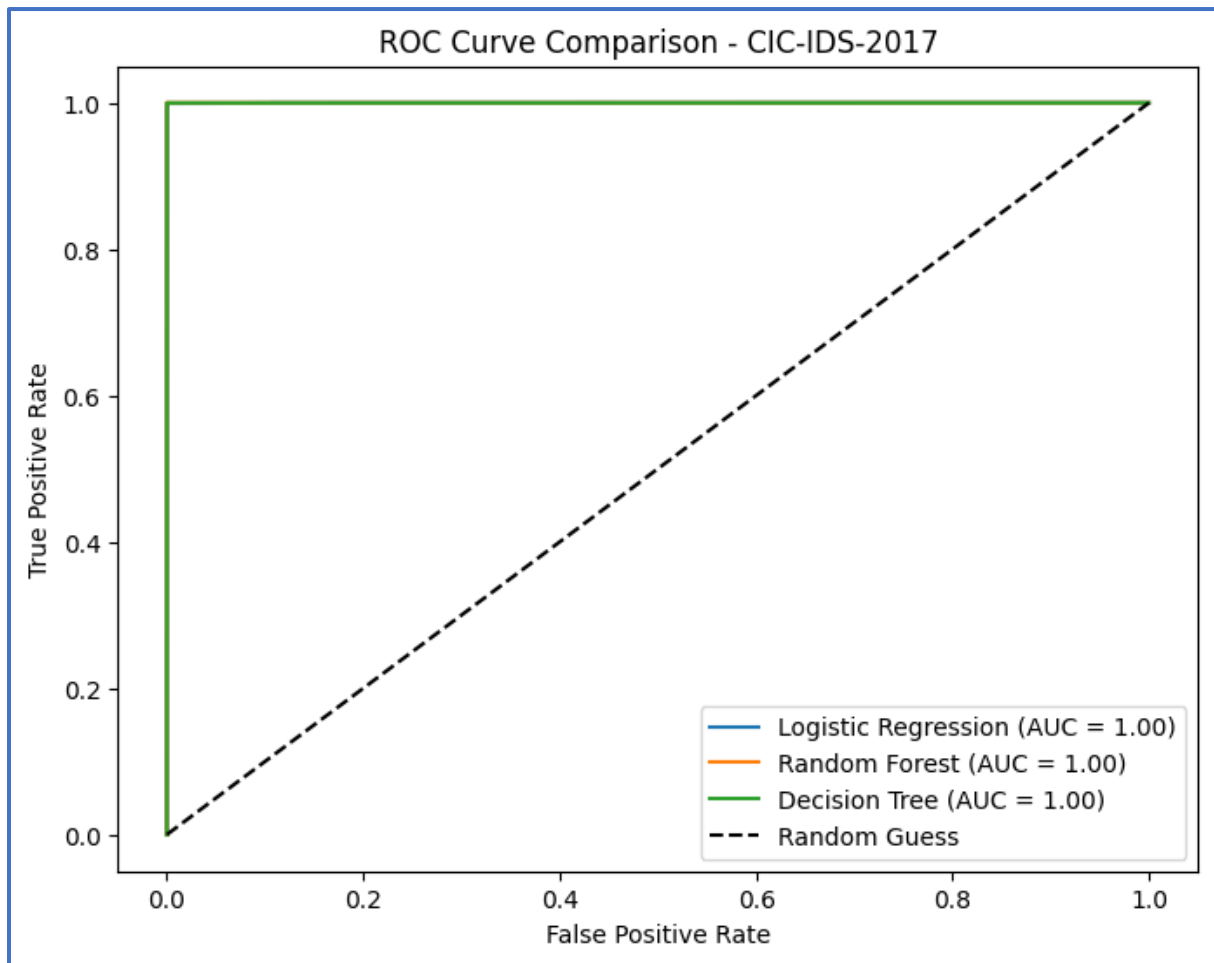


Figure 7: ROC-AUC curves for Logistic Regression, Random Forest, and Decision Tree.

V. Discussion

Sentinel Net demonstrates that AI-powered NIDS can significantly enhance network security. The system's high accuracy and balanced precision-recall metrics indicate its reliability in detecting diverse attack types. By analyzing feature importance, critical network attributes can be prioritized, improving interpretability and operational efficiency.

VI. Conclusion

This study confirms that integrating machine learning with CIC-IDS data can produce a highly effective intrusion detection system. Sentinel Net successfully identifies network threats with minimal false positives, making it a valuable tool for cybersecurity defense.

VII. Future Works

Future research will expand Sentinel Net to other subsets of CIC-IDS-2017, including DDoS and Brute Force attacks, to test generalization. Deep learning models such as CNNs and LSTMs may capture temporal flow patterns. Real-time IDS deployment, dimensionality reduction, and ensemble deep models remain promising directions for scalability and efficiency.